6. Practical - Hamming Code

AIM: Write a Program to implement error detection and correction using HAMMING Code concept. Make a test run to input data stream and verify error correction feature.

Error correction at Data Link Layer:

Hamming code is a set of error-correction codes can be used to detect and correct the errors.

Create Sender Program with below feature

1. Input to sender file should be a text of any length. Program should convert the text to binary.
2. Apply hamming code concept on the binary data and add redundant bits to it
3. Save this output in a file called channel.

Create a receiver Program with below feature

1. Receiver Program should read the input from channel files.
2. Apply hamming code on the binary data to check for errors.
3. If there is an error, display the position of the error.
4. Else remove the redundant bits and convert the binary data to ascii and display the output.



```
data = input("Enter binary data: ")
n = len(data)
r = 0
while 2 ** r < n + r + 1:
    r += 1
m = n + r
h = [0] * m
j = 0
for i in range(1, m + 1):
    if i & (i - 1) != 0:
        h[i - 1] = int(data[j])
        j += 1
for i in range(r):
    p = 2 ** i
    parity = 0
    for j in range(1, m + 1):
        if j & p:
            parity ^= h[j - 1]
    h[p - 1] = parity
print("Hamming Code:", h)
pos = int(input("Enter error position (0 for no error): "))
if 1 <= pos <= m:
    h[pos - 1] ^= 1
print("Received Code:", h)
```

```
CodeRed - powershell.exe
Windows PowerShell
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PS C:\Users\Lenovo\OneDrive\Desktop\CodeRed> python Hamming_Code.py
Enter binary data: 1011
Hamming Code: [0, 1, 1, 0, 0, 1, 1]
Enter error position (0 for no error): 5
Received Code: [0, 1, 1, 0, 1, 1, 1]
Error at position: 5
Corrected Code: [0, 1, 1, 0, 0, 1, 1]
PS C:\Users\Lenovo\OneDrive\Desktop\CodeRed>
```

Hamming code:

```
data = input("Enter binary data: ")
m = len(data)
# calculate number of parity bits
r = 0
while 2**r < m + r + 1:
    r += 1
n = m + r
h = [0] * n
# Insert data bits
j = 0
for i in range(1, n+1):
    if i & (i-1) != 0:
        h[i-1] = int(data[j])
        j += 1
# calculate Parity bits
for i in range(r):
    P = 2**i
    Parity = 0
    for j in range(1, n+1):
        if j & P:
            Parity ^= h[j-1]
    h[P-1] = Parity
print("Hamming code", h)
# Introduce an error
pos = int(input("Enter error position (0 for
    no error): "))
if 1 <= pos <= n:
    h[pos-1] ^= 1
```

```
    Print ("Received code", h)
# Detect error
error = 0
for i in range (r):
    p = 2 ** i
    Parity = 0
if Parity:
    error += p
# Correct error
if error == 0:
    Print ("No error detected")
elif error <= n:
    Print ("Error at position", error)
    h [error - 1] ^= 1
else:
    Print ("Error cannot be corrected")
    Print ("Corrected code", h)
```

Write this in the code section.

Result :-

Thus the Hamming code is used to detect and correct single bit error.

Practical-7 Sliding Window

AIM: Write a Program to implement flow control at data link Layer using SLIDING WINDOW PROTOCOL simulate the flow of frames from one node to another.

Create a Sender Program with following features :-

1. Input window size from the user
2. Input a Text message from the user.
3. Consider 1 character Per Frame.
4. Created a frame with following fields
5. Send the frames.
6. wait for the acknowledgement from the Receiver. [Induce delay in the Program].
7. Reader a file called Receiver-Buffer.
8. check Ack field for the Acknowledgement number.
9. If the Acknowledgement number is a expected Send new set of frames accordingly. [over write the Sender-Buffer file with new frames] [ELSE if Nack is received, resend the frames accordingly.

Create a receiver file with following features.

1. Reader a file called Sender-Buffer.
2. check the Frame no.
3. If the fame no. are as expected.